

# XXXXXXXXXXXXX

xxxxx@gmail.com | (xxx) xxx-xxxx | xxxx@ucsd.edu

## EDUCATION

**University of California, San Diego**

2024 - 2028

B.S. Computer Engineering, Jacobs School of Engineering

La Jolla, CA

GPA: xxxx/4.00

**California Academy of Mathematics and Science**

Aug 2020 - Jun 2024

Weighted GPA: xxx | Unweighted GPA: xxx

Carson, CA

## PROJECTS

**SIMO LED Driver**

Aug 2026 - Present

- Designed a three-channel single-inductor multiple-output (SIMO) boost LED driver with independent regulation and PWM dimming, progressing from LTspice simulation to KiCad PCB layout.
- Implemented peak-current control with NAND gates, Schmitt-trigger inverters, and a cross-coupled SR latch to sequence inductor charging and discharging without a dedicated controller IC.
- Modeled package and PCB parasitic inductance; tuned gate resistance to reduce simulated peak MOSFET drain-to-source voltage from 100 V to 68 V while evaluating switching losses and dead time.

**Analog PWM Speed Controller**

May 2026

- Built an analog PWM speed controller from ECE lab components, using cascaded RC filters to convert a servo PWM input to a control voltage.
- Generated a 3 kHz carrier with a CD4069 inverter oscillator and RC filter; used an LM311 comparator to produce variable-duty-cycle PWM.

## TECHNICAL SKILLS

**Design:** LTspice, PSpice, KiCad

**Test equipment:** Oscilloscope, multimeter

**Programming:** Python, Java, C, MATLAB, C++

## INTERESTS

RC car repair (batteries, speed controllers, motors); bicycle maintenance; photography.